Multimodal Emotion Recognition

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Motivation

- Multiple modalities -> richer representation -> more accurate inference and expression.

- Each modality is expected to provide unique information,
  - Redundancy.

- Challenges (D'Melo and Kory, 2012):
  - Classifiers and fusion methods that better capture the relationships between different modalities.
  - Affective corpora that contains adequate samples of synchronized expressions.
  - Improvements are still often relatively modest.

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Multimodal Affect Recognition

- Underlying relationships and correlation between feature sets in different modalities and affect dimensions
  - Feature Selection
- How different affective expressions influence each other.
  - Uniqueness
- How much information each of them provides about the expressed affect.
  - Modality Selection

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Data Collection

- The need for corpora and databases that contain spontaneous and subtle.

- Databases containing contextual descriptions *synchronized* with other modalities.

- Affective stimuli should be labeled simultaneously for all the modalities available to the coder (e.g. audio + video) rather than considering the single modalities (e.g. audio/video) separately.

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Feature Extraction

- Variable sampling frequency
  - Video processing: 25 FPS | GSR: 16 Hz | EEG: 128 Hz etc.

- Synchronization is required
  - To unite the feature information or
  - To come to a decision at a certain moment in time

- Feature Selection: Optimize the feature space individually per information stream followed by a combined feature selection (Schuller et al., 2008).
  - I. Often highly correlated information should be reduced individually per modality
  - II. Secondary optimization process can lead to further improvements reducing cross-modal redundancy

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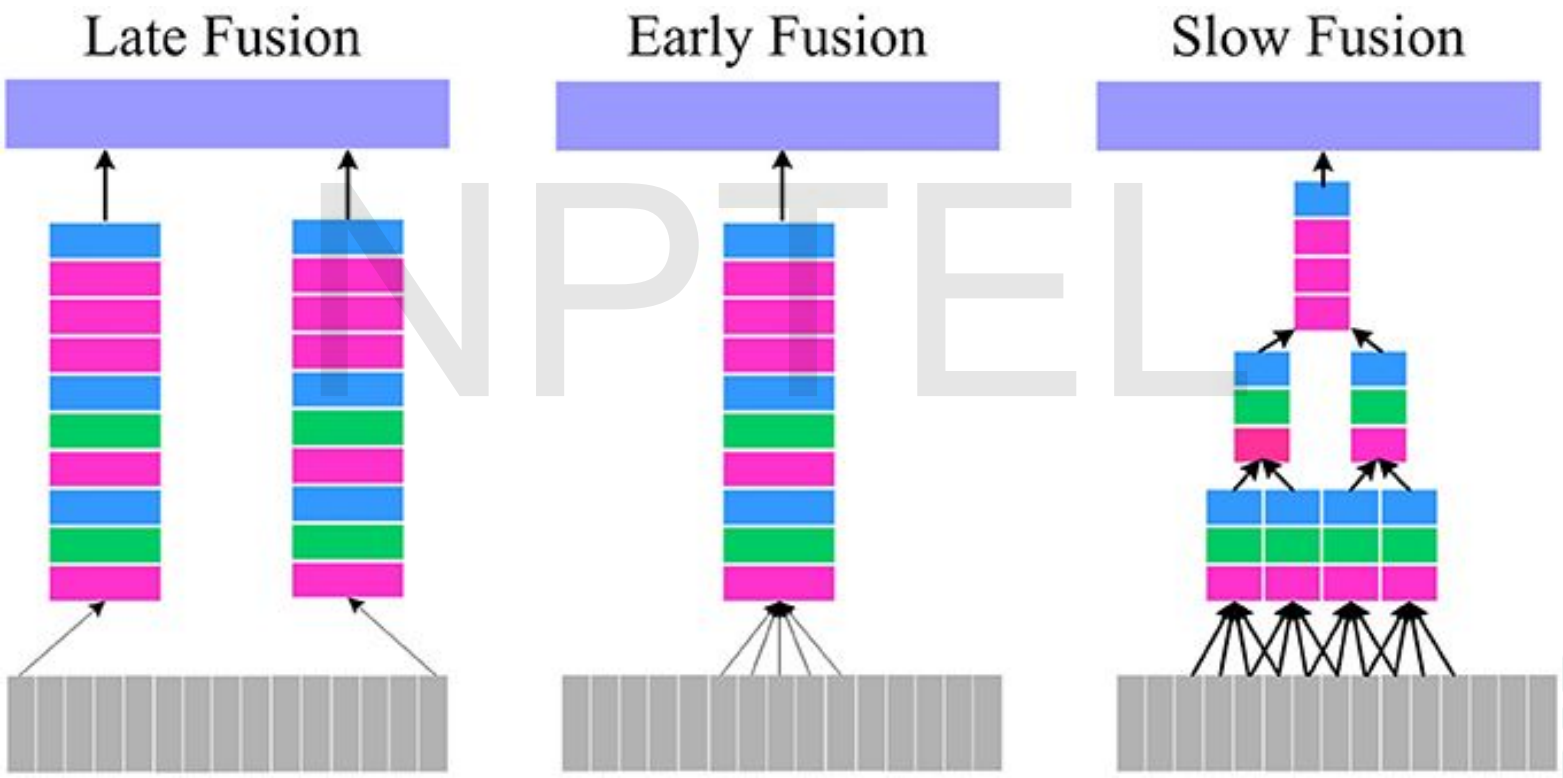
Stream Fusion



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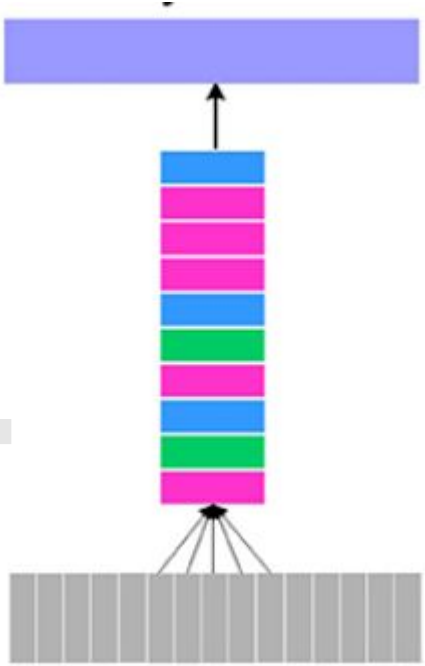
Early Fusion

- Concatenating the the features from multiple cues into one feature vector.
- Becomes more challenging as the number of features increases and when the features are of very different nature.
- Synchronization is of utmost importance.



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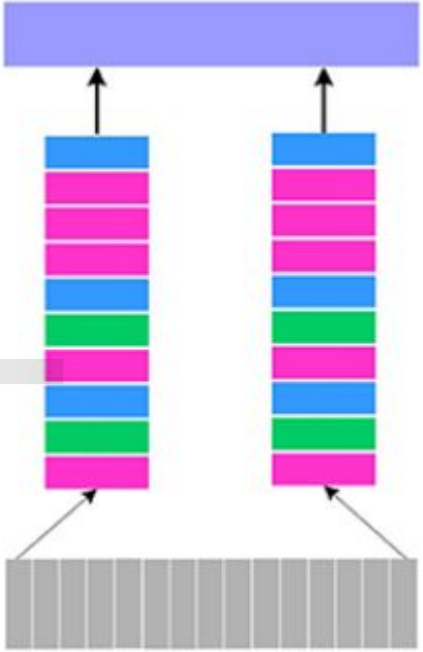
Late Fusion

- Feature and time dependency are abstracted.
- Each classifier process its own data stream and the multiple sets of outputs are combined at a later stage to produce the final results.
- Soft level: a measure of confidence is associated with the decision
- Hard level: the combining mechanism operates on single hypothesis decisions.



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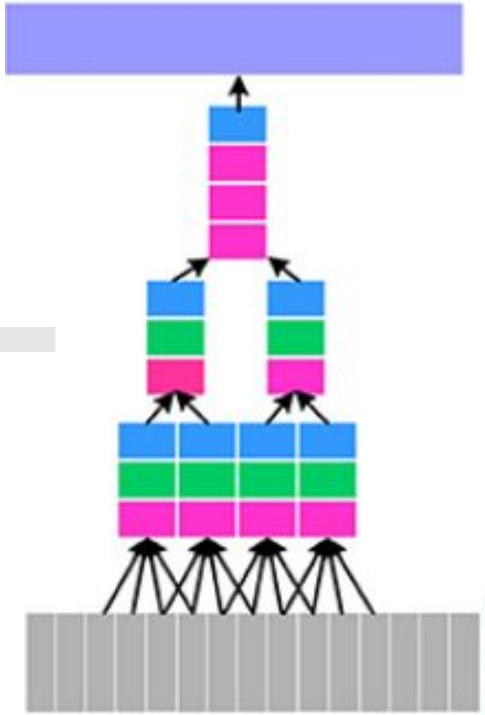
Slow Fusion

- Assumption of conditional independence between modalities and cues in decision level fusion can result in loss of information.

- Exploit the correlations between the modalities while relaxing the requirement of the synchronization.



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SEMAINE

- Sensitive Artificial Listener (SAL): A multimodal dialogue system with the social interaction skills needed for a sustained conversation with a human user.
- Aims to engage the user in a dialog and create an emotional workout by paying attention to the user's non-verbal expressions and reacting accordingly.
  - E.g. Nodding and smiling.

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Multimodal Affect Recognition

- Real-time Multimodal Data
  - Audio-visual data obtained from cameras and microphones
- Data segmentation and analysis
  - Frame/Millisecond/Second/Minute level
  - Trade-off
    - Real-time prediction requires smaller window
    - Reliable prediction accuracy requires longer-term monitoring.
  - Appropriate unit of analysis is context dependent
- Video was recorded at 49.979 fps at a spatial resolution of 780 x 580 pixels and 8 bits per sample, while audio was recorded at 48 kHz with 24 bits per sample.
- Audio-video signals were synchronised with an accuracy of 25 µs.

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SEMAINE Database

- Spontaneous data capturing the audiovisual interaction between a human and an operator undertaking the role of an avatar with four personalities:
  - Prudence, who is even tempered and sensible;
  - Poppy, who is happy and outgoing;
  - Spike, who is angry and confrontational; and
  - Obadiah, who is sad and depressive.
- All interactions were annotated by 2 to 8 raters in the four dimensions in continuous time and continuous value.
- Dimensions are Arousal, Expectation, Power, and Valence.
  - Four dimensions account for most of the distinctions between everyday emotion categories (Fontaine, J., et. al., 2007).
- $\{\mathbf{v}^{a}_{i}, \mathbf{v}^{e}_{i}, \mathbf{v}^{p}_{i}, \mathbf{v}^{v}_{i}\}$ for every rater i and dimension a/e/p/v.

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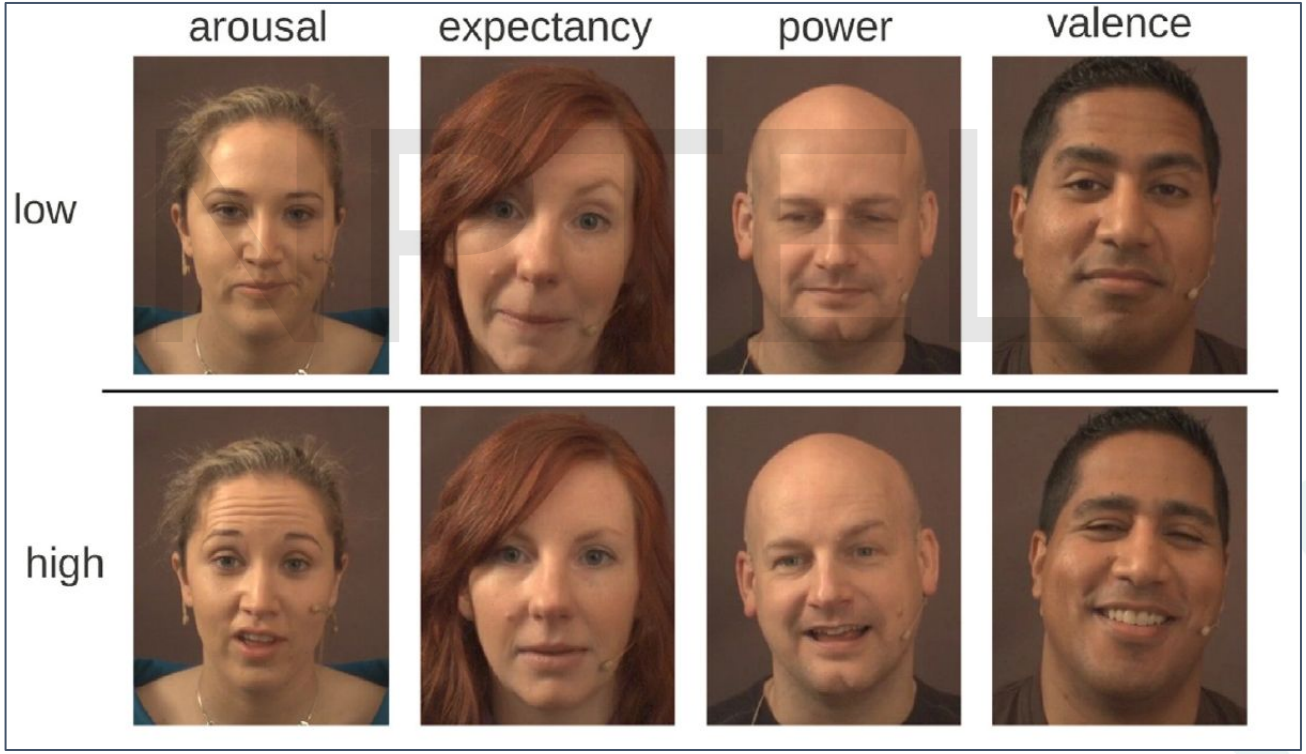
Audio-Visual Emotion Challenge

Uni and multi-modal challenge based on acoustic, linguistic and video cues



Data and Annotations

- To attain binary labels,
  - Average value of each dimension over all raters was computed, resulting in a set of continuous time, real valued variables $\{\mathbf{v^a}, \mathbf{v^e}, \mathbf{v^p}, \mathbf{v^v}\}$
  - Mean of these average ratings over all interactions in the dataset was computed, resulting in the scalar values: $\{\mu^a_i, \mu^e_i, \mu^p_i, \mu^v_i\}$
  - The binary labels $\{\mathbf{y^a}, \mathbf{y^e}, \mathbf{y^p}, \mathbf{y^v}\} \in \{\pm 1\}$ are then found by thresholding $\mathbf{v}^j_t > \mu^j$ for each dimension j at every frame t.
- Video stream was chunked in two ways:
  - Frame level: Per frame for the video only taks
  - Word level: Per word for the audio and audio-visual tasks

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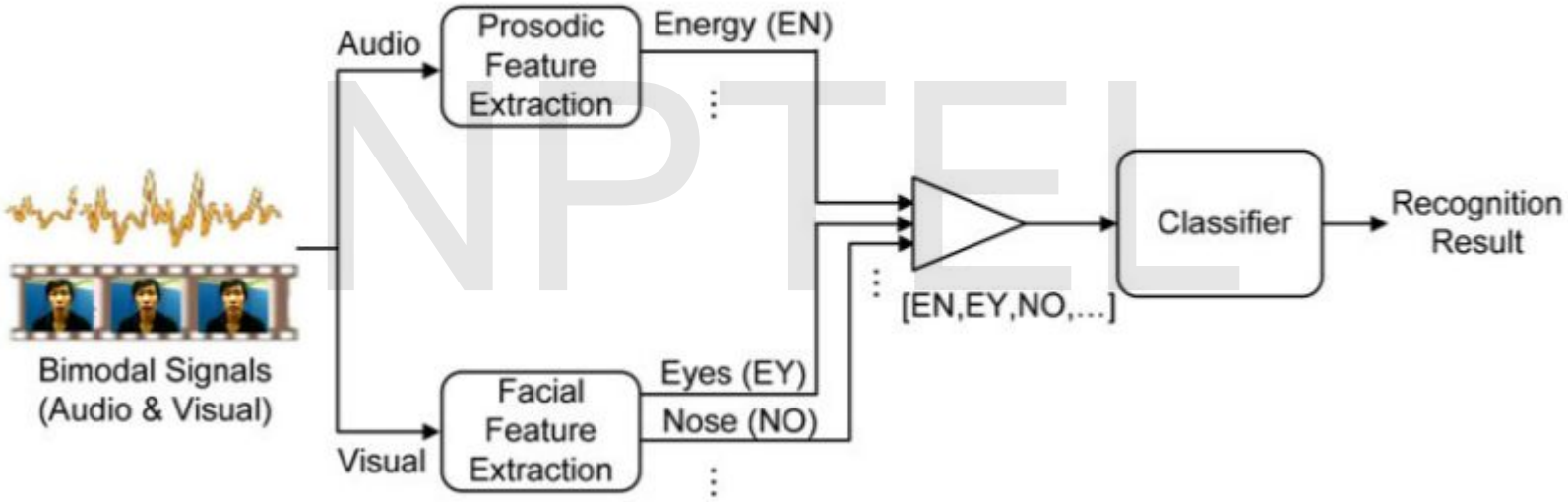
Feature-level fusion



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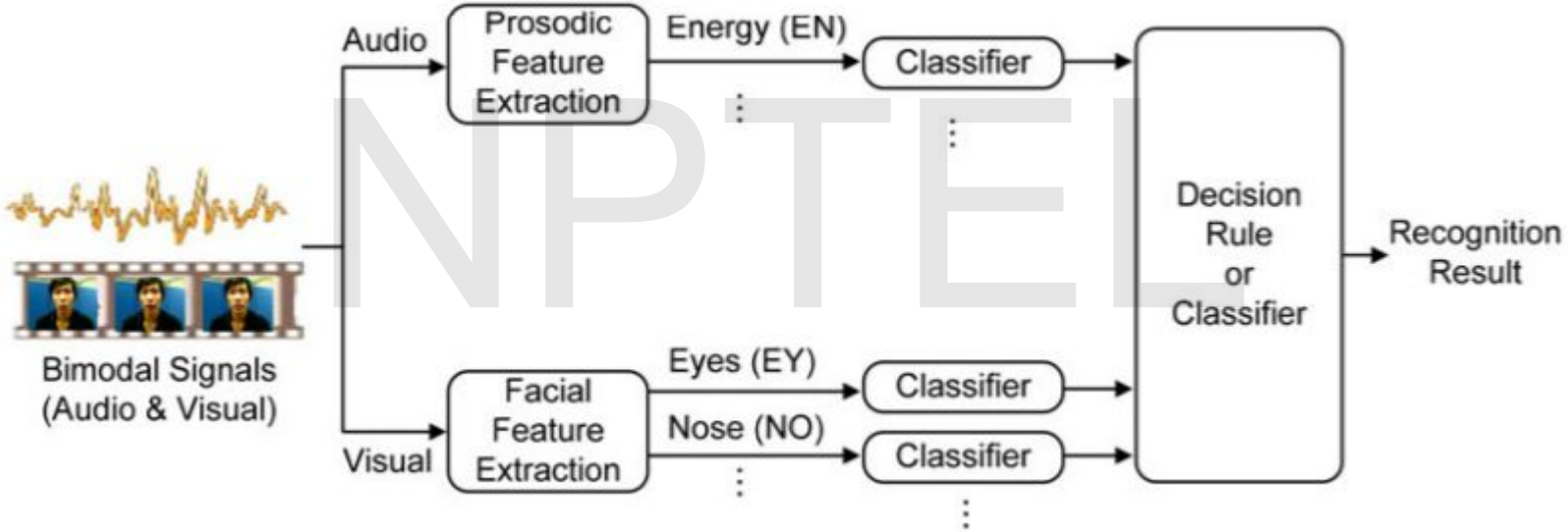
Decision-level fusion



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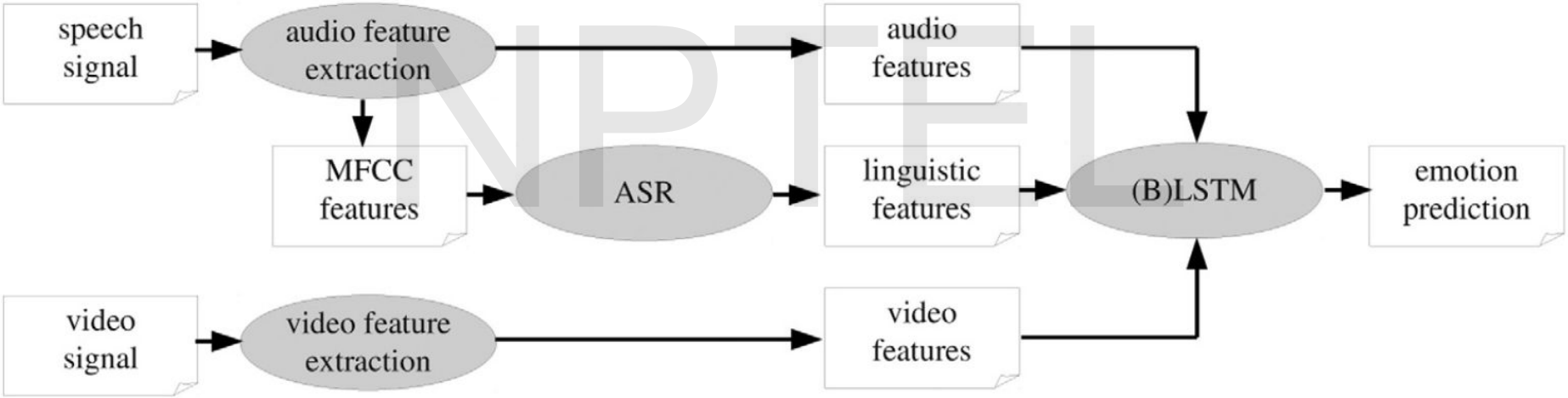
Audio/visual emotion challenge (Wöllmer, M., 2013)



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Results: No Feature Selection

Classifier	Features	AROUSAL		EXPECTATION		POWER		VALENCE		Mean
		WA	UA	WA	UA	WA	UA	WA	UA	WA
BLSTM	A	**68.5**	69.3	64.3	53.5	66.1	53.3	66.3	56.1	66.3
BLSTM	A+L	67.8	69.0	64.8	52.0	65.5	53.9	66.3	56.2	66.1
LSTM	A	**68.5**	68.6	66.1	55.9	64.7	56.1	65.6	55.2	66.2
LSTM	A+L	68.2	68.8	65.2	51.9	**66.2**	55.0	63.8	55.9	65.9
SVM [6]	A	63.7	64.0	63.2	52.7	65.6	55.8	58.1	52.9	62.7
BLSTM	V	62.3	62.9	62.3	51.8	55.2	53.0	63.3	60.5	60.8
LSTM	V	60.3	61.3	60.4	57.7	57.0	50.4	64.0	57.9	60.4
SVM [6]	V	60.2	57.9	58.3	56.7	56.0	52.8	63.6	60.9	59.5
BLSTM	A+V	67.7	68.0	63.1	53.4	60.6	55.0	67.2	61.8	64.7
BLSTM	A+L+V	66.9	67.0	66.2	57.3	63.4	52.3	65.9	61.5	65.6
LSTM	A+V	68.0	67.5	65.7	57.7	63.8	54.7	65.5	59.5	65.8
LSTM	A+L+V	67.4	66.8	65.3	56.7	61.7	54.2	67.6	62.8	65.5
BLSTM (LF)	A+V	67.9	69.3	65.0	53.2	64.0	55.5	**69.8**	61.3	**66.7**
BLSTM (LF)	A+L+V	67.0	68.6	65.7	51.6	63.6	55.7	**69.8**	61.2	66.5
LSTM (LF)	A+V	62.6	64.3	**67.6**	57.6	65.1	56.0	68.2	57.7	65.9
LSTM (LF)	A+L+V	66.3	67.4	63.9	58.1	66.0	53.9	66.4	58.2	65.7

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Discussion: No FS

- For arousal, the best WA of 68.5 % is obtained for acoustic features only, showing that audio is the most important modality for assessing arousal (M. Wollmer, et. al., 2010).
- The classification of expectation seems to benefit from including visual information as the best WA (67.6 %).
- Power is best classified via speech-based features. However, for unidirectional modeling WA significantly increases from 64.7 % to 66.2 % when using linguistics in addition to audio features
- For valence, the inclusion of video information helps, leading to a WA of 69.8 %.

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Results: CFS Feature Selection

Classifier	Features	AROUSAL		EXPECTATION		POWER		VALENCE		Mean
		WA	UA	WA	UA	WA	UA	WA	UA	WA
BLSTM	A	71.3	70.2	66.2	51.0	66.0	56.4	65.9	60.6	67.4
BLSTM	A+L	**73.7**	74.4	66.1	53.1	64.6	55.7	65.8	57.2	67.6
LSTM	A	70.4	69.8	**67.7**	54.6	64.9	58.8	63.1	55.3	66.5
LSTM	A+L	71.9	71.1	63.1	55.5	66.6	56.3	64.7	56.9	66.6
BLSTM	V	59.8	58.8	66.2	50.1	64.1	57.5	63.3	56.0	63.4
LSTM	V	62.7	61.5	66.0	50.1	**70.2**	62.4	64.3	52.7	65.8
BLSTM	A+V	67.8	69.5	64.3	52.3	60.1	57.0	64.7	58.8	64.2
BLSTM	A+L+V	69.9	70.7	63.3	50.4	61.9	56.1	61.4	55.9	64.1
LSTM	A+V	69.7	70.8	64.5	52.0	63.5	56.8	62.4	53.0	65.0
LSTM	A+L+V	70.4	71.3	65.7	53.3	63.5	55.9	62.9	53.2	65.6
BLSTM (LF)	A+V	68.5	67.5	66.7	50.4	64.2	52.7	**69.1**	60.6	67.1
BLSTM (LF)	A+L+V	72.3	72.3	66.6	50.9	64.4	54.0	67.9	58.5	**67.8**
LSTM (LF)	A+V	65.7	63.7	67.4	52.1	68.0	58.6	66.8	54.8	67.0
LSTM (LF)	A+L+V	64.8	63.5	67.1	54.9	68.1	57.3	65.7	56.4	66.4

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Discussion: FS

- For most settings, CFS does not significantly improve the average weighted accuracy.
- However, for recognition based on video only, CFS leads to a remarkable performance gain, increasing the mean WA from 60.4 % to 65.8 % for unidirectional LSTM networks.

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Audio/visual emotion challenge

classifier	features	AROUSAL		EXPECTATION		POWER		VALENCE		mean
		WA	UA	**WA**	UA	**WA**	UA	**WA**	UA	**WA**
BLSTM	A	69.2	69.1	63.1	54.6	59.6	52.9	68.7	57.4	**65.2**
LSTM	A	**71.2**	71.2	57.6	48.7	57.4	50.4	68.7	59.5	63.7
SVM [30]	A	59.8	59.7	63.6	50.0	57.9	48.4	**70.2**	54.9	62.9
ELM-NN [31]	A	52.0	52.3	63.7	50.1	62.2	50.7	69.1	50.0	61.8
AdaBoost [32]	A	57.6	57.5	62.2	49.6	54.2	47.9	60.3	47.6	58.6
LDCRF [33]	A	60.9	60.4	53.2	44.1	56.8	45.7	60.9	45.8	57.9
GMM [34]	A	55.3	55.2	56.1	50.7	49.1	45.3	50.9	48.4	52.9
BLSTM	V	43.1	42.9	**68.6**	62.0	44.8	41.0	51.7	52.4	52.1
LSTM	V	48.6	48.7	65.6	60.2	37.6	35.8	60.8	52.2	53.1
SVM [52]	V	47.8	47.4	62.0	54.8	57.9	47.4	69.6	50.2	59.3
LDCRF [33]	V	53.2	53.1	46.8	43.2	57.3	50.5	59.3	50.7	54.1
BLSTM	A+V	58.3	58.1	64.1	59.5	46.9	45.4	51.1	45.4	55.1
BLSTM	A+L+V	58.8	58.6	60.8	54.8	46.9	44.0	57.1	50.2	55.9
LSTM	A+V	56.3	56.2	61.6	54.1	46.7	45.8	61.2	53.9	56.5
LSTM	A+L+V	57.9	57.8	64.0	58.6	47.6	44.8	55.7	47.9	56.3
SVM [6]	A+V	67.2	67.2	36.3	48.5	62.2	50.0	66.0	49.2	57.9
LDCRF [33]	A+V	65.6	65.3	53.4	49.2	**62.9**	58.3	59.5	49.6	60.3
MLP [35]	A+V	54.1	54.3	58.5	57.8	42.7	40.0	44.8	35.9	50.0
BLSTM (LF)	A+V	69.5	69.4	63.6	54.5	55.8	49.3	69.6	59.2	64.6
BLSTM (LF)	A+L+V	63.3	63.2	62.9	53.0	53.1	48.4	57.6	46.9	59.2
LSTM (LF)	A+V	67.8	67.8	58.3	48.9	57.5	50.3	68.7	59.3	63.1
LSTM (LF)	A+L+V	70.3	70.3	62.5	52.2	56.0	50.4	69.2	58.5	64.5

Discussion: AVEC Challenge

- Audio features lead to the best result for arousal classification.
- For classification of expectation using facial movement features, the obtained WA of 68.6 % is higher than what is reported for other techniques.
- For power, audiovisual classification with Latent-Dynamic Conditional Random Fields as proposed in (G. Ramirez, et. al., 2011) outperformed.
- For valence, the audio features lead to the highest accuracy (70.2 %) (A. Sayedelahl, et. al., 2011).

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